

TOPS TECH Task 2 (CS)

Task -2 Types of Malwares Viruses

1. **List 5 real-world Indicators of Compromise (IoCs) that you might find on a system infected by malware, and for each, explain in one line how it could be detected in system or network logs.**

Ans:

Indicators of Compromise (IoCs) and Their Detection in System and Network Logs

Objective

- The objective of this task is to understand the concept of Indicators of Compromise (IoCs) and identify common signs that indicate a computer system or network has been compromised.
- IoCs help cybersecurity professionals detect malware infections, unauthorized access, suspicious network activities, and other security incidents by analyzing system logs, network logs, and security monitoring tools.

Introduction

- An Indicator of Compromise (IoC) is a piece of digital evidence that suggests a system, application, or network may have been compromised by malicious activity.
- IoCs are commonly used by Security Operations Centers (SOCs), incident response teams, and cybersecurity analysts to identify attacks, investigate security incidents, and respond quickly before significant damage occurs.
- IoCs can be found in various sources such as operating system logs, firewall logs, authentication logs, antivirus reports, DNS logs, web server logs, and network traffic analysis.
- By continuously monitoring these indicators, organizations can detect cyber threats at an early stage and improve their overall security posture.

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Five Real-World Indicators of Compromise (IoCs)

Indicator of Compromise (IoC)	Description	How It Can Be Detected in System or Network Logs
Multiple Failed Login Attempts	Repeated unsuccessful login attempts may indicate a brute-force attack against user accounts.	Analyze authentication or security logs to identify repeated failed login events from the same IP address within a short period.
Suspicious IP Address Connections	Connections from unknown or blacklisted IP addresses may indicate malicious communication or unauthorized access.	Monitor firewall logs, IDS/IPS logs, and network traffic logs to identify connections from suspicious or unauthorized IP addresses.
Unusual Network Traffic	An unexpected increase in network traffic or communication with unknown external servers may indicate malware or data exfiltration.	Analyze network monitoring logs, NetFlow records, or firewall logs to detect abnormal traffic patterns and unusual outbound connections.
Unexpected File Modifications	Critical system or application files are modified without authorization, indicating possible malware infection or unauthorized access.	Monitor file integrity logs, endpoint security alerts, and operating system event logs for unexpected file creation, deletion, or modification activities.
Malicious Process Execution	Unknown or suspicious processes running on the system may indicate malware execution or unauthorized software installation.	Review system event logs, endpoint detection logs, or process monitoring tools to identify unknown executable files and abnormal process behavior.

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Importance of Indicators of Compromise

Indicators of Compromise play a significant role in cybersecurity because they help security teams:

- Detect cyberattacks at an early stage.
- Identify malware infections.
- Monitor suspicious user activities.
- Detect brute-force login attempts.
- Identify unauthorized access to systems.
- Investigate security incidents effectively.
- Reduce the impact of cyber threats.
- Improve incident response capabilities.

Benefits of Monitoring IoCs

Continuous monitoring of IoCs provides several advantages to organizations:

- Early detection of cyber threats.
- Faster incident response.
- Improved network visibility.
- Better protection against malware and ransomware.
- Reduced financial and operational losses.
- Enhanced compliance with cybersecurity standards.
- Increased overall system security.

- ⇒ Indicators of Compromise (IoCs) are essential elements of modern cybersecurity operations.
- ⇒ They provide valuable evidence that helps identify malicious activities within systems and networks.
- ⇒ Common IoCs such as failed login attempts, suspicious IP addresses, unusual network traffic, unexpected file modifications, and malicious process execution enable security analysts to detect attacks at an early stage.

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⇒ Continuous monitoring of system logs and network logs using security information and event management (SIEM) tools such as Splunk or ELK Stack significantly improves an organization's ability to respond to cyber threats and protect critical assets.

2. **Install Splunk (or use Splunk Free Cloud Trial), upload a sample web server log file, and create a dashboard that shows the top 5 IP addresses accessing the server.**

Ans:

Install Splunk, Upload a Sample Web Server Log File, and Create a Dashboard Showing the Top 5 IP Addresses Accessing the Server

Objective

- The objective of this practical is to learn how to use Splunk as a Security Information and Event Management (SIEM) tool for log analysis.
- In this task, a sample web server log file is uploaded into Splunk, and a dashboard is created to display the top five IP addresses accessing the web server.
- This practical demonstrates how security analysts monitor server activity and identify unusual access patterns using log data.

Introduction

- Splunk is a powerful SIEM platform that collects, indexes, and analyzes machine-generated data from different sources such as web servers, operating systems, firewalls, applications, and network devices.
- It enables cybersecurity professionals to monitor security events, detect suspicious activities, generate dashboards, and perform real-time log analysis.
- In this practical, Apache web server logs are analyzed to identify the IP addresses generating the highest number of requests.

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Observation

- The Apache web server log file was successfully uploaded into Splunk.
- Using the Search & Reporting application, the uploaded log data was analyzed to identify the most active client IP addresses.
- A dashboard was created to visualize the top five IP addresses accessing the web server.
- The graphical representation made it easier to identify frequently occurring IP addresses and monitor server access patterns.

Result

- The sample web server log file was successfully imported into Splunk.
- The search query identified the top five client IP addresses generating requests to the web server.
- The results were visualized using a dashboard, demonstrating how Splunk can be used for effective log monitoring, traffic analysis, and security event investigation.

Conclusion

- Splunk provides a powerful platform for centralized log collection, analysis, and visualization.
- By uploading web server logs and creating dashboards, security analysts can quickly identify high-frequency client IP addresses, detect unusual access patterns, and improve overall security monitoring.
- This practical demonstrated the use of Splunk as an effective SIEM solution for web server log analysis.

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3. **Using ELK Stack (Elasticsearch, Logstash, Kibana) or Splunk, write a query to detect failed login attempts from the same IP address within a 10-minute window.**
Hint:
Use aggregation or stats functions to group by IP and filter by event type.

Ans:

Detect Failed Login Attempts from the Same IP Address Within a 10-Minute Window Using Splunk

Objective

- The objective of this practical is to detect multiple failed login attempts originating from the same IP address within a 10-minute time window using Splunk.
- Detecting repeated failed login attempts is an important security monitoring activity because it helps identify brute-force attacks and unauthorized access attempts.
- Splunk's Search Processing Language (SPL) allows security analysts to efficiently analyze authentication logs and detect suspicious login patterns.

Introduction

- Failed login attempts are one of the most common Indicators of Compromise (IoCs) observed during cybersecurity monitoring.
- Attackers often use automated tools to perform brute-force attacks by repeatedly trying different passwords against a user account.
- If several failed login attempts originate from the same IP address within a short period, it may indicate an attempted compromise.
- Splunk provides powerful search and aggregation capabilities that allow security analysts to detect these events quickly and generate alerts before attackers gain unauthorized access.

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Observation

- The authentication logs were analyzed using Splunk's Search Processing Language (SPL).
- The query grouped failed login events based on the source IP address and a 10-minute time interval.
- The analysis successfully identified IP addresses generating multiple failed login attempts, indicating possible brute-force attacks or unauthorized access attempts.

Result

- The Splunk search successfully detected repeated failed login attempts from the same IP address within a 10-minute time window.
- This practical demonstrated how SIEM tools can identify brute-force attacks by analyzing authentication logs and aggregating login events.

Conclusion

- Monitoring failed login attempts is a critical cybersecurity practice for detecting brute-force attacks and unauthorized access attempts.
- Splunk provides powerful log analysis and aggregation features that enable security analysts to quickly identify suspicious authentication activities.
- Implementing such monitoring improves an organization's security posture and supports timely incident response.

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4. **Using Simulate a suspicious activity by inserting a fake log entry that shows an unusual port scan (e.g., multiple connection attempts to different ports from one IP), and use Splunk or Kibana to identify and visualize this anomaly.**

Ans:

Simulate a Suspicious Port Scan Activity and Detect It Using Splunk

Objective

- The objective of this practical is to simulate a suspicious network activity by creating fake log entries representing a port scanning attack and analyze these logs using Splunk.
- The goal is to identify unusual connection attempts from the same IP address to multiple destination ports and visualize this activity using Splunk dashboards.
- This practical demonstrates how Security Information and Event Management (SIEM) tools help detect reconnaissance attacks before an actual system compromise occurs.

Introduction

- Port scanning is one of the most common reconnaissance techniques used by attackers before launching a cyberattack.
- During a port scan, an attacker attempts to connect to multiple ports on a target system to discover open services and identify potential vulnerabilities.
- Security analysts monitor firewall logs, IDS/IPS logs, and server logs to detect repeated connection attempts from a single IP address targeting multiple ports within a short period.
- Splunk enables organizations to collect, analyze, and visualize these events, making it easier to detect suspicious activities and respond quickly.

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Observation

- The uploaded log file showed repeated connection attempts from the same source IP address (192.168.1.100) to multiple destination ports within a short time interval.
- This behavior is commonly associated with a port scanning attack. Splunk successfully grouped the events and displayed the targeted ports, making the suspicious activity easy to identify.

Analysis

- The analysis revealed that the same source IP attempted to access several commonly used service ports within a very short period.
- Such behavior is abnormal for regular users and is a strong indicator of reconnaissance activity.
- Port scanning is often performed before exploitation to identify open services and potential vulnerabilities on a target system.
- Splunk's search and visualization capabilities helped detect this suspicious pattern efficiently, allowing security analysts to investigate the event and take preventive actions such as blocking the offending IP address or generating an automated security alert.

Result

- The simulated port scanning activity was successfully detected using Splunk.
- The log analysis identified multiple denied connection attempts from a single IP address targeting different ports.
- The dashboard clearly visualized the suspicious behavior, demonstrating how SIEM solutions assist in identifying reconnaissance attacks.

Conclusion

- Port scanning is an early-stage attack technique frequently used by cybercriminals to gather information about target systems.
- Monitoring firewall and network logs with Splunk enables organizations to detect such suspicious activities before attackers exploit vulnerabilities.